

2016 Fall ICLAB Midterm Test

1.

Please **only** draw the combinational circuit of the following code

A.) (6%)

assign y = x | c | d;

assign x = a & b;

B.) (9%)

```
module MidtermSoEasy(out,a,b,sel);
```

```
input a, b, sel;
```

```
output out;
```

```
wire seln;
```

```
wire w0, w1;
```

```
not #(0.2:0.3:0.5, 0.2:0.3:0.5) n0( seln, sel );
```

```
and #(0.4:0.6:0.9, 0.4:0.6:0.9) a0( w0, a, seln );
```

```
and #(0.4:0.6:0.9, 0.4:0.6:0.9) a1( w1, b, sel );
```

```
or #(0.4:0.6:0.9, 0.4:0.6:0.9) o0( out, w0, w1 );
```

```
endmodule
```



2.

A.) Please explain setup time and hold time? (%6)

3.

A.) Please simply describe the difference of display, monitor and strobe. (簡答題) (5%)

B.) Please list at least 3 differences between "task" and "function". (簡答題) (5%)

C.) Timescale is very important in hardware design, so it is necessary to set the right timescale while designing your architecture. Please look at the following setting: (5%)

```
`timescale 1ns/XXX
```

```
module TEST
```

```
parameter CYCLE=2.5;
```

```
reg CLK;
```

```
initial CLK=1;
```

```
always #(CYCLE/2.0) CLK=~CLK;
```

```
endmodule
```

What is the minimum value of XXX for precision requirement?

4.

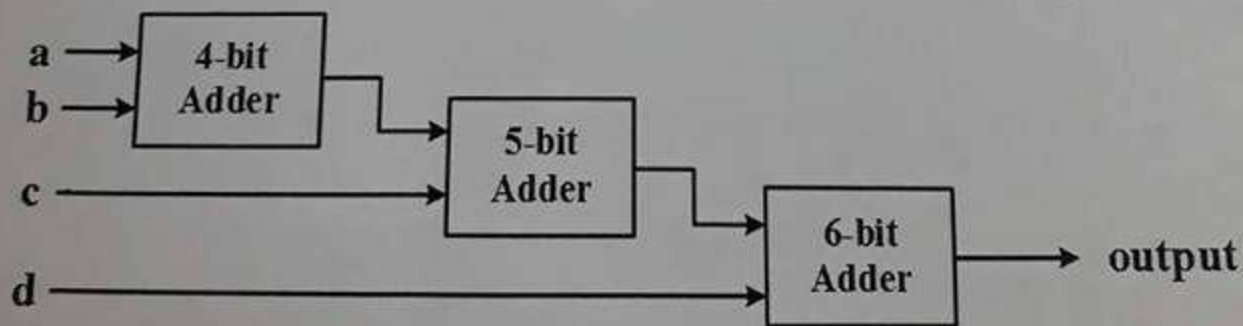
A.) Please calculate the maximum t_{pd} such that this circuit can satisfy the setup time margin. (4%)

B.) Please calculate the minimum t_{cd} such that this circuit can satisfy the hold time margin. (4%)

Note: You should use the following table to express your answer or you can't get any score.

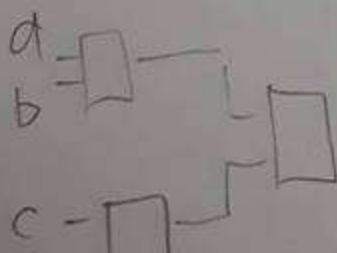
t_{ccq}	Clk-to-Q contamination delay
t_{pcq}	Clk-to-Q propagation delay
t_{cd}	Logic Contamination Delay
t_{pd}	Logic Propagation Delay
t_{skew}	Difference between CLK_1 and CLK_2
t_{setup}	Setup time of DFF
t_{hold}	Hold time of DFF
T_{cycle}	Clock period

C.) a, b, c and d are 4-bit number. Please modify the following circuit to parallel architecture and three-stage pipelined architecture. For doing $(a+b+c+d)$ four times, calculate their area and consuming time respectively. For 1-bit adder, its area is 1 and its delay time is 1. (6%)



$$(a+b) + (b+c)$$

$$00 \quad 0 \quad w$$



$$a+b+c+d$$

5. (10%)

Please describe the difference between three macro.

category	format	description

6. (15%)

A.) What is the sdf file? Why do we use it in gate-level simulation?

B.) Is the following timing report "MET"? Please calculate the slack value for this timing report. If the slack is negative, you should list two ways to solve this problem.

Point	Incr	Path
clock CLK (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
COUNT_LOAD_reg[1] CK (DFFX1)	0.00	0.00 r
COUNT_LOAD_reg[1] QN (DFFX1)	0.56	0.56 r
U134/Y (NAND3X1)	0.23	0.79 f
U160/Y (INVX2)	0.90	1.69 r
U137/Y (MUX2X2)	0.52	2.21 f
.....		
U226/Y (OR4X4)	0.57	8.24 f
U206/Y (NOR2X4)	0.97	9.21 r
U149/Y (AOE2X1)	0.27	9.49 f
U148/Y (INVX2)	0.32	9.80 r
ADDR_reg[0] D (EDFFX1)	0.00	9.80 r
data arrival time	9.80	
clock CLK (rise edge)	10.00	10.00
clock network delay (ideal)	0.00	10.00
ADDR_reg[0] CK (EDFFX1)	0.00	10.00 r
library setup time	-0.29	9.71
data required time	9.71	